Chapter 3

Vector spaces and Subspaces

Homework List-4

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Section 3.1

Spaces of vectors

What is a vector space?

Definition:

- **A vector space is a set of vectors such that all linear combinations of vectors from the set are also in the set.**

Terminology:

- **A mathematical set is simply a defined group of objects in which order does not matter. We generally group objects of a set by curly braces:** $S = \{1,2,3\} = \{3,2,1\} = \{2,1,3\}$.

What is a vector space?

Definition in mathematical notation:

A set of vectors, V, is a vector space if and only if:

A set of vectors, V, is a vector space if and only if:

1. If $\vec{v} \in V$, then $k\vec{v} \in V$, for any scalar k.
2. If $\vec{u}, \vec{v} \in V$, then $\vec{u} + \vec{v} \in V$.

Corollaries:

3. If $\vec{u}, \vec{v} \in V$, then $k\vec{u} + l\vec{v} \in V$, for all scalars k, l; this follows from (1) and (2).
4. The zero vector must be in the set: $\vec{0} \in V$; this follows from (1) with $k = 0$.

Notation: We read the expression, $\vec{v} \in V$, as "$\vec{v}$ is an element of V" or "$\vec{v}$ is in V". This simply means that the vector $\vec{v}$ is a member of the set V.

Geometric interpretation in 2-D

- **For vector in 2-D, the graph of all linear combinations forms a single line which passes through the origin. It also generates a line for any number of vectors that are all on the same line.**

- **For any two or more vectors that are not on the same line (i.e. one line is not a scalar multiple of the other, or independent vectors), all linear combinations yield the entire 2-D space, $\mathbb{R}^2$.**

Geometric interpretation in 3-D

- **For any one vector in 3-D, all linear combinations (i.e. scalar multiples or dependent vectors) lie on a single line which passes through the origin.**
- **For any two vectors that are not on the same line (i.e., independent vectors), all linear combinations lie on a plane which passes through the origin.**
- **For any three vectors are not all on the same plane (independent), all linear combinations yield the entire 3-D space, $\mathbb{R}^3$.**

Geometric interpretation in n dimensions

- **A vector space is a "flat" geometric object that passes through the origin.**
  - **A line through the origin is a vector space.**
  - **A plane through the origin is a vector space.**
  - **A hyperplane through the origin is a vector space.**
- **An entire *n* - D space, $\mathbb{R}^n$, is a special vector space.**
- **The origin, $V_0 = \left\{\vec{0}\right\}$, is also a vector space and a special one.**

Subspaces

A vector space, U, is a subspace of another, V, if (and only if):

- **Every element of U is also an element of V; $\vec{u} \in V, \forall \vec{u} \in U$.**
- **U is also a vector space; $\vec{u}, \vec{v} \in U$, then $k\vec{u} + l\vec{v} \in U$.**

Geometrically:

- **A line (through the origin) in 2-D is a subspace of $\mathbb{R}^2$.**
- **A line (through the origin) in 3-D is a subspace of $\mathbb{R}^3$.**
- **A plane (through the origin) in 3-D is a subspace of $\mathbb{R}^3$.**
- **The 4-D origin, $\left\{\vec{0} = (0,0,0,0)\right\}$, is a subspace of $\mathbb{R}^4$.**

We often refer to $\mathbb{R}^2$ or $\mathbb{R}^3$ (or $\mathbb{R}^n$) as the "whole space" or the "parent space." Formally, $\mathbb{R}^n$ is a subspace of itself.

Linear spans.

- **One way to define a vector space is by providing a spanning set of a (usually small) number of vectors.**
- **The linear span of this set is the set of all possible linear combinations of the vectors of the spanning set; it is a vector space (easy to check by definition).**

$$S = \{(1, \quad 0, \quad 1), (1, \quad 0, \quad -1)\}$$

$$\begin{aligned} U = \ span(S) \quad &= \{k(1,0,1) + l(1,0,-1), \forall k, l \in \mathbb{R}\} \\ &= \{((k+l), \quad 0, \quad (k-l)) \ \forall k, l \in \mathbb{R}\} \\ &= \{(a, \quad 0, \quad b) \ \forall a, b \in \mathbb{R}\} \end{aligned}$$

$$T = \{(1, \quad 2, \quad 3)\}$$

$$\begin{aligned} V = \ span(T) \quad &= \{k(1,2,3), \forall k \in \mathbb{R}\} \\ &= \{(k, \quad 2k, \quad 3k) \ \forall k \in \mathbb{R}\} \end{aligned}$$

Abstract vector spaces.

- **Vectors have two well-defined operations: addition and scalar multiplication.**
  - **In vector addition, we add corresponding terms. Vector addition is associative, commutative and has both an identity and an inverse:**

$$\vec{u} + (\vec{v} + \vec{w}) = (\vec{u} + \vec{v}) + \vec{w}, \qquad \vec{u} + \vec{v} = \vec{v} + \vec{u},$$
$$\exists \vec{0}: \vec{u} + \vec{0} = \vec{u}, \forall \vec{u}, \qquad \forall \vec{u}, \exists -\vec{u}: \vec{u} + (-\vec{u}) = \vec{0}$$

  - **In scalar multiplication, we multiply all elements of the vector by the same value. Scalar multiplication is distributive with vector addition and is compatible with "regular" scalar multiplication:**

$$k(\vec{u} + \vec{v}) = k\vec{u} + k\vec{v}, \qquad (k + l)\vec{u} = k\vec{u} + l\vec{u},$$
$$(kl)\vec{u} = k(l\vec{u}), \qquad 1\vec{u} = \vec{u}, \forall \vec{u}$$

- **Definition: Vector spaces are called closed with respect to these operations:** $\vec{u} + \vec{v} \in V, \forall \vec{u}, \vec{v} \in V, \qquad k\vec{v} \in V, \forall \vec{v} \in V, \forall k \in \mathbb{R}$

Definition: Vector spaces.

- **If we can define a set of mathematical objects, V, that:**
  - **The set has two well-defined operations, addition and scalar multiplication:**
  - **where vector addition is associative, commutative and has both an identity and an inverse:**

$$\boldsymbol{u}+(\boldsymbol{v}+\boldsymbol{w})=(\boldsymbol{u}+\boldsymbol{v})+\boldsymbol{w}, \qquad \boldsymbol{u}+\boldsymbol{v}=\boldsymbol{v}+\boldsymbol{u},$$
$$\exists \mathbf{0} \in V: \boldsymbol{u}+\mathbf{0}=\boldsymbol{u}, \forall \boldsymbol{u} \in V, \qquad \forall \boldsymbol{u} \in V, \exists -\boldsymbol{u} \in V: \boldsymbol{u}+(-\boldsymbol{u})=\mathbf{0}$$

  - **scalar multiplication is distributive with vector addition and is compatible with "regular" scalar multiplication:**

$$k(\boldsymbol{u}+\boldsymbol{v})=k\boldsymbol{u}+k\boldsymbol{v}, \qquad (k+l)\boldsymbol{u}=k\boldsymbol{u}+l\boldsymbol{u}$$
$$(kl)\boldsymbol{u}=k\,(l\boldsymbol{u}) \qquad 1\boldsymbol{u}=\boldsymbol{u}, \forall \boldsymbol{u} \in V$$

  - **and that is closed with respect to these operations:**

$$\boldsymbol{u}+\boldsymbol{v} \in V, \forall \boldsymbol{u}, \boldsymbol{v} \in V, \qquad k\boldsymbol{u} \in V, \forall \boldsymbol{u} \in V, \forall k \in \mathbb{R}$$

- **We call V a "vector space", and the elements of V are "vectors".**

Matrix space.

- **Consider the set of all $m \times n$ matrices $\mathbb{R}^{m\times n}$.**
- **We can add matrices to get another matrix of the same size**

$$A + B \in \mathbb{R}^{m\times n}, \forall A, B \in \mathbb{R}^{m\times n}$$

$$A + (B + C) = (A + B) + \mathrm{C}, \qquad A + B = B + A,$$

- **We can define a zero matrix and an additive inverse:**

$$\exists 0 \in \mathbb{R}^{m\times n}: A + 0 = A, \forall A \in \mathbb{R}^{m\times n}, \qquad \forall A \in \mathbb{R}^{m\times n}, \exists -A \in \mathbb{R}^{m\times n}: A + (-A) = 0$$

- **We can multiply matrices by a scalar to get another matrix of the same size**

$$kA \in \mathbb{R}^{m\times n}, \forall A \in \mathbb{R}^{m\times n}, \forall k \in \mathbb{R}$$

$$k(A + B) = kA + kB, \qquad (k + l)A = kA + lA$$

$$(kl)A = k\,(lA), \qquad 1A = A, \forall A \in \mathbb{R}^{m\times n}$$

- **The set of all matrices of the same size is a vector space!**

Function space.

- **Consider the set of all functions of a single variable, $F = \{f(x)\}$.**
- **We can add functions to get another function of a single variable:**

$$f(x) + g(x) = h(x) \rightarrow h(x) = (f + g)(x)$$

- **We can multiply functions by a scalar to get another function of a single variable:**

$$kf(x) = g(x) \rightarrow g(x) = (kf)(x)$$

- **Function addition: group like terms and add coefficients:**
  - **Obeys all the rules; the zero function is $f(x) = 0$.**
- **Scalar multiplication: multiply all terms by the same scalar:**
  - **Obeys all the rules.**
- **The set of all functions of a single variable is a vector space!**

Polynomial space.

- **Consider the set of all functions of a single variable that can be written as a polynomial:**

$$P = \{ f(x) \mid f(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n \}$$

- **When we add polynomials, we get another polynomial:**

$$f(x) + g(x) \in P, \forall f(x), g(x) \in P$$

- **When we scale polynomials, we get another polynomial:**

$$kf(x) \in P, \forall f(x) \in P, \forall k \in \mathbb{R}$$

- **The set of polynomials is closed with respect to addition and scalar multiplication.**
- **The set of all polynomial functions of a single variable is a vector space; it is a subspace of function space!**

Column Space

The column space and solvable right-hand sides.

Linear spans of rows and columns

- An m x n matrix can be thought of either as:
  - m row vectors, each of length n, $A = \begin{bmatrix} (\vec{a}_1^R)^T \\ (\vec{a}_2^R)^T \\ \vdots \\ (\vec{a}_m^R)^T \end{bmatrix}$
  - n column vectors, each of length m, $A = \begin{bmatrix} (\vec{a}_1^C) & (\vec{a}_2^C) & \cdots & (\vec{a}_n^C) \end{bmatrix}$
- We can take the linear span of these sets of vectors!

The row space of a matrix.

- **The row space of a matrix is the linear span of the vectors corresponding to each row.**
- **We define a spanning set from the rows:**

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} = \begin{bmatrix} (\vec{a}_1^R)^T \\ (\vec{a}_2^R)^T \\ \vdots \\ (\vec{a}_m^R)^T \end{bmatrix}, (\vec{a}_i^R)^T = [a_{i1} \quad a_{i2} \quad \cdots \quad a_{in}]$$

$$S_R = \{\vec{a}_1^R, \vec{a}_2^R, \cdots, \vec{a}_m^R\}$$

- **and this gives us the row space:**

$$\boldsymbol{R}(A) = rowsp(A) = span(S_R) = span(\{\vec{a}_1^R, \vec{a}_2^R, \cdots, \vec{a}_m^R\})$$

The column space of a matrix.

- **The column space of a matrix is the linear span of the vectors corresponding to each column.**
- **We define a spanning set from the columns:**

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} = \begin{bmatrix} (\vec{a}_1^C) & (\vec{a}_2^C) & \cdots & (\vec{a}_n^C) \end{bmatrix}, (\vec{a}_j^C) = \begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}$$

$$S_C = \{\vec{a}_1^C, \vec{a}_2^C, \cdots, \vec{a}_n^C\}$$

- **and this gives us the column space:**

$$\boldsymbol{C}(A) = colsp(A) = span(S_C) = span(\{\vec{a}_1^C, \vec{a}_2^C, \cdots, \vec{a}_n^C\})$$

The column picture of a linear system.

- **A linear system can be viewed as a linear combination of columns.**

$$A\vec{x} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix}$$

$$A\vec{x} = \vec{b} \rightarrow x_1(\vec{a}_1^C) + x_2(\vec{a}_2^C) + \cdots + x_n(\vec{a}_n^C) = \vec{b}$$

- **A linear system $A\vec{x} = \vec{b}$ has an exact solution if and only if $\vec{b}$ can be written as a linear combination of the columns of A.**

The meaning of the column space

- A linear system $A\vec{x} = \vec{b}$ has a solution if and only if $\vec{b}$ can be written as a linear combination of the columns of A.

$$A\vec{x} = \vec{b} \rightarrow x_1(\vec{a}_1^C) + x_2(\vec{a}_2^C) + \cdots + x_n(\vec{a}_n^C) = \vec{b}$$

- The column space of A is the set of all possible linear combinations of the columns of A.

$$\boldsymbol{C}(A) = colsp(A) = span(\{\vec{a}_1^C, \vec{a}_2^C, \cdots, \vec{a}_n^C\})$$

- The column space of A is the set of all possible right-hand sides for which $A\vec{x} = \vec{b}$ has at least one solution.
- A system $A\vec{x} = \vec{b}$ has a solution if and only if $\vec{b}$ is in the column space of A.
- The column space of A is a subspace of $\mathbb{R}^m$, where m is the number of rows in A.
  - This includes the possibility of the column space being $\mathbb{R}^m$ itself.
  - This includes the possibility of the column space being just the zero vector, but only for one very special matrix.

Possible column spaces in 2-D.

- **If each column of a 2-by-n matrix is a scalar multiple of each other, then the column space is a line in 2-D.**

$$A = \begin{bmatrix} 2 & -4 & -2 \\ 1 & -2 & -1 \end{bmatrix} \rightarrow x\begin{bmatrix} 2 \\ 1 \end{bmatrix} + y\begin{bmatrix} -4 \\ -2 \end{bmatrix} + z\begin{bmatrix} -2 \\ -1 \end{bmatrix} = (x - 2y - z)\begin{bmatrix} 2 \\ 1 \end{bmatrix} = k\begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

- **If there are (at least) two columns of a 2-by-n matrix that are not scalar multiples of each other, the column space is the whole $\mathbb{R}^2$.**

$$A = \begin{bmatrix} 2 & 3 \\ 1 & -1 \end{bmatrix} \rightarrow x\begin{bmatrix} 2 \\ 1 \end{bmatrix} + y\begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

Possible column spaces in 3-D.

- **If each column of a 3-by-n matrix is a scalar multiple of each other, the column space is a line in 3-D.**
- **If the columns of a 3-by-n matrix can all be expressed as linear combinations of only two of the columns (and these are not all scalar multiples of each other), the column space is a plane in 3-D.**

$$\begin{bmatrix} 1 & 2 & 2 & 3 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \to w\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + x\begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} + y\begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix} + z\begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix} = (w + 2x + z)\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + (y + z)\begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix} = k\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + l\begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}$$

- **If there are (at least) three columns of a 3-by-n matrix that can not be written as linear combinations of each other of each other, the column space is all of $\mathbb{R}^3$.**

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \to x\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + y\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + z\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Example 1:

Describe the column spaces (they are subspaces of $\mathbf{R}^2$) for

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 4 \end{bmatrix}.$$

- The column space of I is the entire space R^2.
- The column space of A is only a line because the second column is a multiple of the first column.
- The column space of B is all of R^2 because not all vectors are multiples of each other.

Example 2:

- Question: Keep only the vectors (x, y) whose components are positive or zero (quarter plane). Is this a subspace?

- Answer: No, you cannot multiply c=-1.

Example 3:

Describe a subspace S of each vector space V.

- V= all symmetric 2 by 2 matrices (a subspace of M)

Answer: A subspace S can always be itself V or zero subspace (2 by 2 matrix with 0 in each entry).

- V= all solutions to the equation $\frac{\mathrm{d}^4 y}{dx^4} = 0$

Answer: A subspace could be all quadratic equations.

Section 3.2

The null space of A and the complete solution to Ax=b.

Homogeneous systems.

- A homogeneous linear system of equations is simply one with a right-hand side of all zeros:

$$A\vec{x} = \vec{0}$$

- Every matrix A can be associated with a (unique) homogeneous system.
- In the row picture of a system, a homogenous system consists of planes (or hyperplanes) that all pass through the origin.
- A homogenous system always has a solution of ; $\vec{x} = \vec{0}$; the trivial solution, a homogeneous system may have other solutions as well (see previous example).

The null space of a matrix.

- **The set of all possible solutions to the homogeneous system, $A\vec{x} = \vec{0}$, is called the null space of A.**

$$\boldsymbol{N}(A) = nullsp(A) = \left\{\vec{x}: A\vec{x} = \vec{0}\right\}$$

- **Any linear combination of any two solutions to $A\vec{x} = \vec{0}$ must also be a solution.**

$$\vec{u}, \vec{v} \in \boldsymbol{N}(A) \rightarrow A\vec{u} = \vec{0}, A\vec{v} = \vec{0}$$
$$\vec{u}, \vec{v} \in \boldsymbol{N}(A) \rightarrow A(k\vec{u} + l\vec{v}) = k(A\vec{u}) + l(A\vec{v}) = \vec{0}$$
$$\vec{u}, \vec{v} \in \boldsymbol{N}(A) \rightarrow k\vec{u} + l\vec{v} \in \boldsymbol{N}(A), \forall k, l \in \mathbb{R}$$

The null space is a vector space!

General solution to a homogeneous system.

- How can we find the null space? → How do we solve a linear system?
- Forward elimination to an echelon form:

$$[A \quad \vec{0}] \rightarrow [U \quad \vec{0}]$$

- Checkpoint: If there are no free variables, there is only one solution.

Trivial solution: $\vec{x} = \vec{0}$

- If there are free variables, we need a general solution (in parametric form).
- Back substitution to the reduced row echelon form:

$$[A \quad \vec{0}] \rightarrow [U \quad \vec{0}] \rightarrow [R \quad \vec{0}]$$

- Solve for the pivot variables in terms of free variables, then substitute a parameter for each free variable (these can take on any value).

Method 1: General solution to a homogeneous system.

- **Example:** $A\vec{x} = \vec{0}$, $A = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 2 & 4 & 2 & 4 \\ 1 & 2 & 3 & 2 \end{bmatrix}$

$$[A \quad \vec{0}] = \begin{bmatrix} 1 & 2 & 1 & 2 & 0 \\ 2 & 4 & 2 & 4 & 0 \\ 1 & 2 & 3 & 2 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 1 & 2 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\to \begin{bmatrix} 1 & 2 & 1 & 2 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 0 & 2 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \to \begin{matrix} x_1 & +2x_2 & +2x_4 & = 0 \\ x_3 & & & = 0 \end{matrix}$$

$$\begin{matrix} x_1 & = & -2x_2 - 2x_4 \\ x_3 & = & 0 \end{matrix} \to (x_1, x_2, x_3, x_4) = (-2s - 2t, s, 0, t), \forall s, t \in \mathbb{R}$$

Note: x_1, x_3 are the pivots, x_2, x_4 are free variables.

Method 1: General solution to a homogeneous system.

- **Alternative form:**

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -2s-2t \\ s \\ 0 \\ t \end{bmatrix} = \begin{bmatrix} -2s \\ s \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} -2t \\ 0 \\ 0 \\ t \end{bmatrix} = s\begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t\begin{bmatrix} -2 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

- **The general solution can be written as a set of all possible linear combination of vectors → a vector space**

Method 2: Special solutions to a homogeneous system.

- **Obtain the general solution to a linear system directly from a set of special solutions.**
- **A special solution has exactly one free variable set equal to 1 and all others set equal to 0.**
- **# of special solutions= # of free variables.**

Method 2: Special solutions to a homogeneous system.

- **Example:** $A\vec{x} = \vec{0}, \quad A = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 2 & 4 & 2 & 4 \\ 1 & 2 & 3 & 2 \end{bmatrix}$

$$[A \quad \vec{0}] = \begin{bmatrix} 1 & 2 & 1 & 2 & 0 \\ 2 & 4 & 2 & 4 & 0 \\ 1 & 2 & 3 & 2 & 0 \end{bmatrix} \rightarrow\rightarrow \begin{bmatrix} 1 & 2 & 0 & 2 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{matrix} x_1 & +2x_2 & & +2x_4 & = 0 \\ x_3 & & & & = 0 \end{matrix}$$

- **2 free variables $x_2, x_4 \longrightarrow$ 2 special solutions:**
  - $x_2 = 1, x_4 = 0 \rightarrow x_1 = -2, x_3 = 0 \rightarrow \ (-2,1,0,0)$
  - $x_2 = 0, x_4 = 1 \rightarrow x_1 = -2, x_3 = 0 \rightarrow \ (-2,0,0,1)$
- **General solution:**

$$(x_1, x_2, x_3, x_4) = s(-2,1,0,0) + t(-2,0,0,1), \forall s, t \in \mathbb{R}$$

Method 3: Reduced row echelon form contains the special solutions.

$$A = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 2 & 4 & 2 & 4 \\ 1 & 2 & 3 & 2 \end{bmatrix} \rightarrow\rightarrow \begin{bmatrix} 1 & 2 & 0 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

- **2 free variables $x_2, x_4 \longrightarrow$ 2 special solutions:**
  - $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} ? \\ 1 \\ ? \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & \mathbf{2} & 0 & 2 \\ 0 & \mathbf{0} & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} \mathbf{-2} \\ 1 \\ \mathbf{0} \\ 0 \end{bmatrix}$
  - $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} ? \\ 0 \\ ? \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 & \mathbf{2} \\ 0 & 0 & 1 & \mathbf{0} \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} \mathbf{-2} \\ 0 \\ \mathbf{0} \\ 1 \end{bmatrix}$

- Set one free variable to 1, others to 0.
- Select corresponding column.
- For each non-zero row:
  - The negative of the element in the free variable column is the element in the special solution corresponding to the pivot of that row.

Method 3: Reduced row echelon form contains the special solutions.

- **An example:** $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} & a_{15} \\ a_{21} & a_{22} & a_{23} & a_{24} & a_{25} \\ a_{31} & a_{32} & a_{33} & a_{34} & a_{35} \\ a_{41} & a_{42} & a_{43} & a_{44} & a_{45} \end{bmatrix} \rightarrow\rightarrow \begin{bmatrix} 1 & 0 & a & 0 & c \\ 0 & 1 & b & 0 & d \\ 0 & 0 & 0 & 1 & e \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} = R$

- **2 free variables $x_3, x_5 \rightarrow$ 2 special solutions:**

Notice: Column 3 is a linear combination of column 1 and 2 in R, then it is also true for A.

$$\vec{s}_1 = \begin{bmatrix} ? \\ ? \\ 1 \\ ? \\ 0 \end{bmatrix} = \begin{bmatrix} -a \\ -b \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad \vec{s}_2 = \begin{bmatrix} ? \\ ? \\ 0 \\ ? \\ 1 \end{bmatrix} = \begin{bmatrix} -c \\ -d \\ 0 \\ -e \\ 1 \end{bmatrix}$$

Find special solutions S_1 and S_2 such that
$RS_1 = 0, RS_2 = 0.$

Then RS=0 implies that AS =0.

Example: possible null spaces in 2-D.

- **If the rows of an m-by-2 matrix are all scalar multiples of each other, there is one free variables and the null space is a line in 2-D. (Special solution!)**

$$A = \begin{bmatrix} 2 & -4 \\ 1 & -2 \\ -3 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & -4 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \boldsymbol{N}(A) = \left\{ \vec{x} : \vec{x} = s\begin{bmatrix} 2 \\ 1 \end{bmatrix}, \forall s \in \mathbb{R} \right\}$$

- **If there are (at least) two rows of an m-by-2 matrix that are not scalar multiples of each other, there are no free variables and the null space is just the zero vector.**

$$A = \begin{bmatrix} 2 & -4 \\ 1 & -3 \\ -2 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & -4 \\ 0 & -1 \\ 0 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}, \boldsymbol{N}(A) = \left\{ \vec{0} \right\}$$

Example: Possible null spaces in 3-D.

- **If the rows of an m-by-3 matrix are all scalar multiples of each other, there are two free variables and the null space is a plane in 3-D. (Special solution!, method 2)**

$$A = \begin{bmatrix} 2 & -4 & 6 \\ 1 & -2 & 3 \end{bmatrix} \to \begin{bmatrix} 2 & -4 & 6 \\ 0 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & -2 & 3 \\ 0 & 0 & 0 \end{bmatrix}, \boldsymbol{N}(A) = \left\{ \vec{x} : \vec{x} = s \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}, \forall s, t \in \mathbb{R} \right\}$$

- **If the rows of an m-by-3 matrix can all be expressed as linear combinations of only two of the rows (and these are not all scalar multiples of each other), there is one free variable and the null space space is a plane in 3-D. (Special solution!, Method 3)**

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & 3 \\ 0 & 1 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \boldsymbol{N}(A) = \left\{ \vec{x} : \vec{x} = s \begin{bmatrix} -2 \\ -1 \\ 1 \end{bmatrix}, \forall s \in \mathbb{R} \right\}$$

- **If there are at least three rows of an m-by-3 matrix that can not be expressed as linear combinations of the others, there are no free variables and the null space is just the zero vector.**

The null space of a matrix.

- **The null space of a matrix is set of solutions to the homogenous system with that coefficient matrix:**

$$\boldsymbol{N}(A) = nullsp(A) = \left\{ \vec{x} : A\vec{x} = \vec{0} \right\}$$

- **The dimension of the null space of A = the number of free variables in an echelon form of A.**
- **The null space of A can be described by a set of special solutions.**
- **The null space of A is a subspace of $\mathbb{R}^n$, where n is the number of columns in A.**
  - **This includes the possibility of the null space being just zero vector or $\mathbb{R}^n$**

Nonhomogeneous systems.

- **For every nonhomogenous system, we can write an associated homogenous system:**

$$A\vec{x} = \vec{b}, \qquad A\vec{x} = \vec{0}$$

- **If $\vec{x}_p$ is any solution to $A\vec{x} = \vec{b}$, and $\vec{x}_n$ is any solution to the homogenous system, then**

$$A(\vec{x}_p + \vec{x}_n) = A\vec{x}_p + A\vec{x}_n = \vec{b} + \vec{0} = \vec{b}$$

Adding any solution of homogenous system to a solution of the nonhomogeneous system will still be a solution to the nonhomogeneous system.

Nonhomogeneous systems.

- If $\vec{x}_{p1}$ and $\vec{x}_{p2}$ are ***both*** solutions to $A\vec{x} = \vec{b}$:

$$A(\vec{x}_{p1} - \vec{x}_{p2}) = A\vec{x}_{p1} - A\vec{x}_{p2} = \vec{b} - \vec{b} = \vec{0}$$

- The difference of any two solutions to the inhomogenous system must be a solution to the homogenous system!
- Every solution to $A\vec{x} = \vec{b}$ is related to every other solution, by the addition of a vector from the null space of A.

$$\forall \vec{x}_{p1}, \vec{x}_{p2}: A\vec{x}_{p1} = A\vec{x}_{p2} = \vec{b}, \exists \vec{x}_n \in N(A): \vec{x}_{p2} = \vec{x}_{p1} + \vec{x}_n$$

- The general solution to $A\vec{x} = \vec{b}$ can be expressed as the sum of one particular solution and the general solution to $A\vec{x} = \vec{0}$.

A geometric interpretation.

- The set of solutions to $A\vec{x} = \vec{0}$ (the null space of A) a flat geometric object passing through the origin.
- The set of solutions to $A\vec{x} = \vec{b}$ is a flat geometric object that does **not** pass through the origin.
- **These two solution sets are parallel objects!**
- The set of solutions to $A\vec{x} = \vec{b}$ is the null space of A, translated away from the origin (it is an affine space, rather than a vector space).
- The general solution to $A\vec{x} = \vec{b}$ is thus $N(A) + \vec{x}_p$ the null space plus a translation vector.

Section 3.3

The complete Solution to $Ax = b$.

General solution to a nonhomogeneous system.

- The general solution to $A\vec{x} = \vec{b}$ can be expressed as the sum of one special solution and the general solution to $A\vec{x} = \vec{0}$.
- Method to find a particular solution: Gaussian Elimination on the augmented matrix!
- Like special solutions to the homogenous system by setting all free variables but one equal to be 0, we can find a special solution by setting every free variable to be zero.
- We can find all the information in the reduced row echelon form of the augmented matrix! So, the key is to find the row echelon form.

General solution to a nonhomogeneous system.

An example: $A\vec{x} = \vec{b}, A = \begin{bmatrix} 1 & 1 & 2 & 2 \\ 2 & 2 & 1 & 1 \\ 3 & 3 & 3 & 3 \\ 1 & 1 & -1 & -1 \end{bmatrix}, \vec{b} = \begin{bmatrix} 0 \\ 2 \\ 2 \\ 0 \end{bmatrix}$

$$[A \quad \vec{b}] = \begin{bmatrix} 1 & 1 & 2 & 2 & 1 \\ 2 & 2 & 1 & 1 & 5 \\ 3 & 3 & 3 & 3 & 6 \\ 1 & 1 & -1 & -1 & 4 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 2 & 2 & 1 \\ 0 & 0 & -3 & -3 & 3 \\ 0 & 0 & -3 & -3 & 3 \\ 0 & 0 & -3 & -3 & 3 \end{bmatrix}$$

$$\to \begin{bmatrix} 1 & 1 & 2 & 2 & 1 \\ 0 & 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 0 & 0 & 3 \\ 0 & 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} = [R \quad \vec{d}]$$

General solution to a nonhomogeneous system.

Pivots: $x_1;\ x_3$
Free variables: $x_2;\ x_4$

$$[R \quad \vec{d}] = \begin{bmatrix} 1 & 1 & 0 & 0 & 3 \\ 0 & 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- **The two free variable columns contain the negative of the pivot values for the two special solutions (Method 3 because it is in the reduced row echelon form). The last column contains the pivot values for the special solution.**

$$\vec{x}_{n1} = \begin{bmatrix} ? \\ 1 \\ ? \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \vec{x}_{n2} = \begin{bmatrix} ? \\ 0 \\ ? \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \end{bmatrix}, \vec{x}_p = \begin{bmatrix} ? \\ 0 \\ ? \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ -1 \\ 0 \end{bmatrix}$$

General solution to a nonhomogeneous system.

An example:

$$A\vec{x} = \vec{b}, A = \begin{bmatrix} 1 & 1 & 2 & 2 \\ 2 & 2 & 1 & 1 \\ 3 & 3 & 3 & 3 \\ 1 & 1 & -1 & -1 \end{bmatrix}, \vec{b} = \begin{bmatrix} 0 \\ 2 \\ 2 \\ 0 \end{bmatrix} \rightarrow \vec{x}_{n1} = \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \vec{x}_{n2} = \begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \end{bmatrix}, \vec{x}_p = \begin{bmatrix} 3 \\ 0 \\ -1 \\ 0 \end{bmatrix}$$

General solution: $S = \{\vec{x} : \vec{x} = \vec{x}_p + s\vec{x}_{n1} + t\vec{x}_{n2} \forall s, t \in \mathbb{R}\}$

$$S = \left\{ \vec{x} : \vec{x} = \begin{bmatrix} 3 \\ 0 \\ -1 \\ 0 \end{bmatrix} + s \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \end{bmatrix} \forall s, t \in \mathbb{R} \right\},$$

The rank of a matrix.

- **Definition: The rank of a matrix is the number of pivots in an echelon form.** $r = \mathrm{rank}(A)$

  The character of the column space and null space are both determined by the number of pivots (in a row echelon form).

  Every free column is a linear combination of earlier pivot columns. (Method 2: special solutions can tell us the combinations).

Properties of matrix with full row rank

1. **All rows have pivots, and R (reduced row echelon matrix) has no zero rows.**
2. $A\vec{x} = \vec{b}$ has a unique **solution for every right-side** $\vec{b}$.
3. The column space is the whole space R^m.
4. There are *n-r = n-m* special solutions in the null space of *A.*

Generalization: Four possible solutions depends on the matrix rank r.

Consider a coefficient matrix with m rows, n columns and a rank of r.

- **Full rank (nonsingular): If $r = m = n$: The matrix is square and non-singular (invertible). Have 1 solution.**
- **Full row rank: If $r = m$, and $n > m$: The matrix is short and wide. Have infinitely many solutions.**
- **Full column rank: If $r = n$, and $m > n$: The matrix is tall and thin. Every column has a pivot. Have 0 or 1 solution.**
- **If $r < m$ and $r < n$, regardless of the values of m and n: Both rows and columns do not contain a pivot. Have 0 or infinitely many solutions.**

Examples on null spaces

General solutions and rank

Example 1: Find the special solutions.

- **Find the special solutions for the following homogenous systems and use the result to describe the null space of the coefficient matrix.**

$$A\vec{x} = \vec{0}, \qquad A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 1 & 2 & 4 \end{bmatrix} \qquad B\vec{x} = \vec{0}, \qquad B = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 3 \\ 1 & 0 & 2 \\ 2 & 2 & 1 \end{bmatrix}$$

$$C\vec{x} = \vec{0}, \qquad C = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 2 & 4 & 2 & 4 \\ 1 & 2 & 3 & 0 \end{bmatrix}$$

Example 1: Case 1

- **Find the special solutions for the following homogenous system and use the result to describe the null space of the coefficient matrix.**

$$A\vec{x} = \vec{0}, \qquad A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 1 & 2 & 4 \end{bmatrix}$$

Example 1: case 1

$$A\vec{x} = \vec{0}, \qquad A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 1 & 2 & 4 \end{bmatrix}$$

- **First, find the reduced row echelon form:**

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 1 & 2 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} = R$$

- **There are one free variables, x_2, and thus one special solution:**

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} ? \\ 1 \\ ? \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} = \vec{x}_n$$

- **The null space is a line:**

$$\boldsymbol{N}(A) = \{\vec{x} : \vec{x} = s\vec{x}_n, \forall s \in \mathbb{R}\}$$

Example 1: case 2

- **Find the special solutions for the following homogenous system and use the result to describe the null space of the coefficient matrix.**

$$B\vec{x} = \vec{0}, \qquad B = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 3 \\ 1 & 0 & 2 \\ 2 & 2 & 1 \end{bmatrix}$$

Example 1: Case 2

$$B\vec{x} = \vec{0}, \qquad B = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 3 \\ 1 & 0 & 2 \\ 2 & 2 & 1 \end{bmatrix}$$

- **First, find the reduced row echelon form:**

$$B = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 3 \\ 1 & 0 & 2 \\ 2 & 2 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & 1 \\ 0 & -2 & 1 \\ 0 & -2 & -1 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \\ 0 & 0 & -3 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\to \begin{bmatrix} 1 & 2 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} = R$$

- **There are no free variables, the null space is the origin, $N(B) = \{\vec{0}\}$, and there are no special solutions.**

Example 1: case 3

- **Find the special solutions for the following homogenous system and use the result to describe the null space of the coefficient matrix.**

$$C\vec{x} = \vec{0}, \qquad C = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 2 & 4 & 2 & 4 \\ 1 & 2 & 3 & 0 \end{bmatrix}$$

Example 1: case 3

$$C\vec{x} = \vec{0}, \qquad C = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 2 & 4 & 2 & 4 \\ 1 & 2 & 3 & 0 \end{bmatrix}$$

- **First, find the reduced row echelon form:**

$$C = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 2 & 4 & 2 & 4 \\ 1 & 2 & 3 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & -2 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 0 & 2 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = R$$

- **There are two free variables, x_2 and x_4, and thus two special solutions:**

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} ? \\ 1 \\ ? \\ 0 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \vec{x}_{n1} \qquad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} ? \\ 0 \\ ? \\ 1 \end{bmatrix} = \begin{bmatrix} -3 \\ 0 \\ 1 \\ 1 \end{bmatrix} = \vec{x}_{n2}$$

- **The null space is a plane:**

$$\boldsymbol{N}(A) = \{\vec{x} : \vec{x} = s\vec{x}_{n1} + t\vec{x}_{n2}, \forall s, t \in \mathbb{R}\}$$

Example 2: Find the general solution.

Find the general solution in the form:

$$\vec{x} = \vec{x}_p + s_1\vec{x}_{n1} + s_2\vec{x}_{n2} + \cdots + s_{(n-r)}\vec{x}_{N,(n-r)}$$

$$(a) \quad \begin{bmatrix} 1 & 1 & 2 & 2 \\ 2 & 2 & 1 & 1 \\ 3 & 3 & 3 & 3 \\ 1 & 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 2 \\ 2 \end{bmatrix}$$

Find the general solution.

$$\begin{bmatrix} 1 & 1 & 2 & 2 \\ 2 & 2 & 1 & 1 \\ 3 & 3 & 3 & 3 \\ 1 & 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 2 \\ 2 \end{bmatrix}$$

- **First, find the reduced row echelon form of the augmented matrix:**

$$[A \quad \vec{b}] = \begin{bmatrix} 1 & 1 & 2 & 2 & 0 \\ 2 & 2 & 1 & 1 & 2 \\ 3 & 3 & 3 & 3 & 2 \\ 1 & 1 & -1 & -1 & 2 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 2 & 2 & 0 \\ 0 & 0 & -3 & -3 & 2 \\ 0 & 0 & -3 & -3 & 2 \\ 0 & 0 & -3 & -3 & 2 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 2 & 2 & 0 \\ 0 & 0 & -3 & -3 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\to \begin{bmatrix} 1 & 1 & 2 & 2 & 0 \\ 0 & 0 & 1 & 1 & -2/3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 0 & 0 & 4/3 \\ 0 & 0 & 1 & 1 & -2/3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \to [R \quad \vec{d}]$$

Find the general solution.

$$\begin{bmatrix} 1 & 1 & 2 & 2 \\ 2 & 2 & 1 & 1 \\ 3 & 3 & 3 & 3 \\ 1 & 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 2 \\ 2 \end{bmatrix}$$

- Next, find the special solutions of $A\vec{x} = \vec{0}$ and a particular solution of $A\vec{x} = \vec{b}$:

$$[R \quad \vec{d}] = \begin{bmatrix} 1 & 1 & 0 & 0 & 4/3 \\ 0 & 0 & 1 & 1 & -2/3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- There are two free variables, x_2 and x_4, and thus two special solutions:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} ? \\ 1 \\ ? \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \vec{x}_{n1} \qquad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} ? \\ 0 \\ ? \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \end{bmatrix} = \vec{x}_{n2} \qquad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} ? \\ 0 \\ ? \\ 0 \end{bmatrix} = \begin{bmatrix} 4/3 \\ 0 \\ -2/3 \\ 0 \end{bmatrix} = \vec{x}_p$$

Find the general solution.

$$\begin{bmatrix} 1 & 1 & 2 & 2 \\ 2 & 2 & 1 & 1 \\ 3 & 3 & 3 & 3 \\ 1 & 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 2 \\ 2 \end{bmatrix}$$

- **Finally, put it all together:**

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} ? \\ 1 \\ ? \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \vec{x}_{n1} \qquad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} ? \\ 0 \\ ? \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \end{bmatrix} = \vec{x}_{n2} \qquad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} ? \\ 0 \\ ? \\ 0 \end{bmatrix} = \begin{bmatrix} 4/3 \\ 0 \\ -2/3 \\ 0 \end{bmatrix} = \vec{x}_p$$

- **The general solution is:**

$$\vec{x} = \vec{x}_p + s\vec{x}_{n1} + t\vec{x}_{n2}, \forall s, t \in \mathbb{R}$$

$$\vec{x} = \begin{bmatrix} 4/3 \\ 0 \\ -2/3 \\ 0 \end{bmatrix} + s \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4/3 - s \\ s \\ -2/3 - t \\ t \end{bmatrix}$$

Example 2: Find the general solution.

Find the general solution in the form:

$$\vec{x} = \vec{x}_p + s_1\vec{x}_{n1} + s_2\vec{x}_{n2} + \cdots + s_{(n-r)}\vec{x}_{N,(n-r)}$$

$$(b)\ \begin{bmatrix} 0 & 1 & 2 & 3 & 0 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 1 & 3 & 4 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

Find the general solution.

$$\begin{bmatrix} 0 & 1 & 2 & 3 & 0 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 1 & 3 & 4 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

- **First, find the reduced row echelon form of the augmented matrix:**

$$[A \quad \vec{b}] = \begin{bmatrix} 0 & 1 & 2 & 3 & 0 & 2 \\ 0 & 0 & 1 & 1 & 2 & -1 \\ 0 & 1 & 3 & 4 & 2 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 & 2 & 3 & 0 & 2 \\ 0 & 0 & 1 & 1 & 2 & -1 \\ 0 & 0 & 1 & 1 & 2 & -1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 0 & 1 & 2 & 3 & 0 & 2 \\ 0 & 0 & 1 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 & 0 & 1 & -4 & 4 \\ 0 & 0 & 1 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow [R \quad \vec{d}]$$

Example 2: case b)

$$\begin{bmatrix} 0 & 1 & 2 & 3 & 0 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 1 & 3 & 4 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

- **Next, find the special solutions of $A\vec{x} = \vec{0}$ and a particular solution of $A\vec{x} = \vec{b}$:**

$$[R \quad \vec{d}] = \begin{bmatrix} 0 & 1 & 0 & 1 & -4 & 4 \\ 0 & 0 & 1 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- **There are three free variables, x_1, x_4 and x_5, and thus three special solutions:**

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 1 \\ ? \\ ? \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \vec{x}_{n1}, \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ ? \\ ? \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ -1 \\ 1 \\ 0 \end{bmatrix} = \vec{x}_{n2}, \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ ? \\ ? \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ -2 \\ 0 \\ 1 \end{bmatrix} = \vec{x}_{n3}, \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ ? \\ ? \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ -1 \\ 0 \\ 0 \end{bmatrix} = \vec{x}_p$$

Example 2: case b)

$$\begin{bmatrix} 0 & 1 & 2 & 3 & 0 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 1 & 3 & 4 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

- **Finally, put it all together:**

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 1 \\ ? \\ ? \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \vec{x}_{n1}, \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ ? \\ ? \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ -1 \\ 1 \\ 0 \end{bmatrix} = \vec{x}_{n2}, \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ ? \\ ? \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ -2 \\ 0 \\ 1 \end{bmatrix} = \vec{x}_{n3},, \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ ? \\ ? \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ -1 \\ 0 \\ 0 \end{bmatrix} = \vec{x}_p$$

- **The general solution is:**

$$\vec{x} = \vec{x}_p + s_1\vec{x}_{n1} + s_2\vec{x}_{n2} + s_3\vec{x}_{n3}, \forall s_i \in \mathbb{R}$$

$$\vec{x} = \begin{bmatrix} 0 \\ 4 \\ -1 \\ 0 \\ 0 \end{bmatrix} + s_1 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + s_2 \begin{bmatrix} 0 \\ -1 \\ -1 \\ 1 \\ 0 \end{bmatrix} + s_3 \begin{bmatrix} 0 \\ 4 \\ -2 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} s_1 \\ 4 - s_2 + 4s_3 \\ -1 - s_2 - 2s_3 \\ s_2 \\ s_3 \end{bmatrix}$$

Example 2: case b)

$$\begin{bmatrix} 0 & 1 & 2 & 3 & 0 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 1 & 3 & 4 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

- **Finally, put it all together:**

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 1 \\ ? \\ ? \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \vec{x}_{n1}, \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ ? \\ ? \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ -1 \\ 1 \\ 0 \end{bmatrix} = \vec{x}_{n2}, \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ ? \\ ? \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ -2 \\ 0 \\ 1 \end{bmatrix} = \vec{x}_{n3},, \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ ? \\ ? \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ -1 \\ 0 \\ 0 \end{bmatrix} = \vec{x}_p$$

- **The general solution is:**

$$\vec{x} = \vec{x}_p + s_1\vec{x}_{n1} + s_2\vec{x}_{n2} + s_3\vec{x}_{n3}, \forall s_i \in \mathbb{R}$$

$$\vec{x} = \begin{bmatrix} 0 \\ 4 \\ -1 \\ 0 \\ 0 \end{bmatrix} + s_1 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + s_2 \begin{bmatrix} 0 \\ -1 \\ -1 \\ 1 \\ 0 \end{bmatrix} + s_3 \begin{bmatrix} 0 \\ 4 \\ -2 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} s_1 \\ 4 - s_2 + 4s_3 \\ -1 - s_2 - 2s_3 \\ s_2 \\ s_3 \end{bmatrix}$$

Example 3: Find the rank.

Find the rank of each of the following matrices.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{bmatrix} \qquad B = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 1 & -1 & 1 \end{bmatrix} \qquad C = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 1 & -1 & 1 \\ 0 & 2 & 4 \end{bmatrix}$$

Example 3: Find the rank.

Reduce to echelon form, and count pivots:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -3 \end{bmatrix} \Rightarrow rank(A) = 2$$

$$B = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 1 & -1 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 1 \\ 0 & -3 & 0 \\ 0 & -3 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 1 \\ 0 & -3 & 0 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow rank(B) = 2$$

$$C = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 1 & -1 & 1 \\ 0 & 2 & 4 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & -2 & 0 \\ 0 & 2 & 4 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 4 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow rank(C) = 3$$

Example 4: Find the rank by inspection

Without performing elimination, identify the rank of each of the following matrices.

$$D = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix} \qquad \mathrm{E} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \qquad \mathrm{F} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Find the rank by inspection

Without performing elimination, identify the rank of each of the following matrices.

$$D = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix} \qquad \mathrm{E} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \qquad \mathrm{F} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- $rank(D) = 1$ **All non-zero rows are identical.**
- $rank(E) = 0$ **All rows are zero.**
- $rank(F) = 4$ **Reduced row echelon form with 4 pivots.**

Example 5:

- Find the condition b_1, b_2, b_3 for the following linear system to have a solution.

$$A\boldsymbol{x} = \boldsymbol{b} \quad \text{is} \quad \begin{aligned} x_1 + 2x_2 + 3x_3 + \ \ 5x_4 &= b_1 \\ 2x_1 + 4x_2 + 8x_3 + 12x_4 &= b_2 \\ 3x_1 + 6x_2 + 7x_3 + 13x_4 &= b_3 \end{aligned}$$

Answer:

$$\begin{bmatrix} 1 & 2 & 3 & 5 & \mathbf{b_1} \\ 2 & 4 & 8 & 12 & \mathbf{b_2} \\ 3 & 6 & 7 & 13 & \mathbf{b_3} \end{bmatrix} \rightarrow \left[\begin{array}{cccc|c} 1 & 2 & 3 & 5 & \mathbf{b_1} \\ 0 & 0 & 2 & 2 & \mathbf{b_2 - 2b_1} \\ 0 & 0 & -2 & -2 & \mathbf{b_3 - 3b_1} \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 2 & 3 & 5 & \mathbf{b_1} \\ 0 & 0 & 2 & 2 & \mathbf{b_2 - 2b_1} \\ 0 & 0 & 0 & 0 & \mathbf{b_3 + b_2 - 5b_1} \end{array}\right]$$